

CM0081 Formal Languages and Automata

§ 9.1 A Language That Is Not Recursively Enumerable

Andrés Sicard-Ramírez

Universidad EAFIT

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Preliminaries

Textbook

John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman (2007). Introduction to Automata Theory, Languages, and Computation.

Conventions

- The number and page numbers assigned to chapters, examples, exercises, figures, quotes, sections and theorems on these slides correspond to the numbers assigned in the textbook.
- The natural numbers include the zero, that is, $\mathbb{N} = \{0, 1, 2, \dots\}$.
- The power set of a set A , that is, the set of its subsets, is denoted by $\mathcal{P}A$.

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Definition

A language L is **undecidable** iff L is not recursive.

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- The term “**recursive**” as synonym for “**decidable**” comes from Mathematics (prior to computers).
- Equivalent formalization to Turing-machine computability based on recursive functions.
- A function is recursive if only if it is Turing-machine computable (see, e.g. (Boolos, Burges and Jeffrey 2007), (Hermes 1969) or (Kleene 1974)).
- Recursive problem: *“it is sufficiently simple that I can write a **recursive function** to solve it, and the function **always** finishes.”* (p. 385)

Codification of Turing Machines

Convention

The Turing machine M is of the form:

$$M = (\{q_1, \dots, q_n\}, \{0, 1\}, \{X_1, X_2, X_3, \dots, X_m\}, \delta, q_1, B, \{q_2\}),$$

where $X_1 = 0$, $X_2 = 1$ and $X_3 = B$. Moreover, $D_1 = L$ and $D_2 = R$.

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Codification of an instruction

The instruction $\delta(q_i, X_j) = (q_k, X_l, D_m)$ is codified by

$$0^i 10^j 10^k 10^l 10^m.$$

Codification of Turing Machines

Codification of a Turing machine

Let $C_1, C_2, \dots, C_p$ be the codifications of the instructions of a Turing machine M . The codification of M is defined by

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Remark

Note that there are other possible codes for M .

Codification of Turing Machines

Enumeration of the binary strings

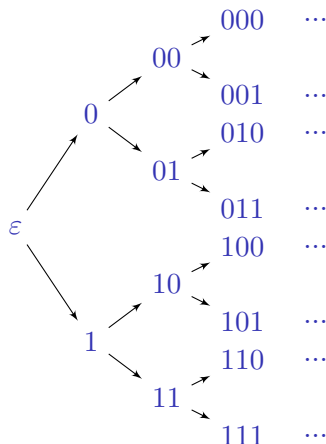
We ordered the binary strings by [length-]lexicographical order (strings are ordered by length, and strings of equal length are ordered lexicographically).

(continued on next slide)

Codification of Turing Machines

Enumeration of the binary strings (continuation)

If w is a binary string, we call w the i -th string where $1w$ is the binary integer i . We refer to the i -th string as w_i .

$$\begin{aligned}\varepsilon &\rightarrow 1_b \rightarrow 1, \\ 0 &\rightarrow 10_b \rightarrow 2, \\ 1 &\rightarrow 11_b \rightarrow 3, \\ 00 &\rightarrow 100_b \rightarrow 4, \\ 01 &\rightarrow 101_b \rightarrow 5, \\ 10 &\rightarrow 110_b \rightarrow 6, \\ &\vdots\end{aligned}$$


Codification of Turing Machines

i -th Turing machine

Given a Turing machine M with code w_i , we can now associate a natural number to it: M is the i -th Turing machine, referred to as M_i .

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Convention

If w_i is not a valid Turing machine code, we shall take M_i to be the Turing machine with one state and no transitions, that is,

$$L(M_i) = \emptyset.$$

Cantor's Diagonalisation Proof

Theorem

The open interval $(0, 1)$ is an uncountable (non-enumerable) set.

(continued on next slide)

Cantor's Diagonalisation Proof

Proof.

Let's suppose $(0, 1)$ is (infinite) countable.

$$r_1 = 0.d_{11}d_{12}d_{13}d_{14} \dots$$

$$r_2 = 0.d_{21}d_{22}d_{23}d_{24} \dots$$

$$r_3 = 0.d_{31}d_{32}d_{33}d_{34} \dots$$

$\vdots$

Let $r = 0.d_1d_2d_3 \dots \in (0, 1)$, where

$$d_i = \begin{cases} 4, & \text{if } d_{ii} \neq 4; \\ 5, & \text{if } d_{ii} = 4. \end{cases}$$

The number r does not belong to the above enumeration. Therefore the interval $(0, 1)$ is an uncountable set. 

The Diagonalization Language

Definition

Let $\Sigma = \{0, 1\}$. The **diagonalization language** is defined by

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		w_j				
		1	2	3	4	...
M_i	1	0	1	1	0	...
	2	1	1	0	1	...
	3	0	1	1	0	...
	4	1	1	0	0	...
	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$

$$a_{ij} = \begin{cases} 1, & \text{if } w_j \in L(M_i); \\ 0, & \text{if } w_j \notin L(M_i). \end{cases}$$

Language $L(M_i)$'s vector: i -th row

L_d : Complement of the diagonal

Is it possible that L_d be in a row?

The Diagonalization Language

Theorem 9.2

The language L_d is not recursively enumerable.

Proof by contradiction (proof of negation)

Whiteboard.

References

- George S. Boolos, John P. Burges and Richard C. Jeffrey [1974] (2007). Computability and Logic. 5th ed. Cambridge University Press (cit. on pp. 6–9).
- Hans Hermes [1961] (1969). Enumerability · Decidability · Computability. Second revised edition. Translated G. T. Hermann and O. Plassmann. Springer-Verlag (cit. on pp. 6–9).
- John E. Hopcroft, Rajeev Motwani and Jefferey D. Ullman [1979] (2007). Introduction to Automata Theory, Languages, and Computation. 3rd ed. Pearson Education (cit. on p. 2).
- Stephen Cole Kleene [1952] (1974). Introduction to Metamathematics. Seventh reprint. Bibliotheca Mathematica: A Series of Monographs on Pure and Applied Mathematics. Volume 1. North-Holland (cit. on pp. 6–9).